

AI Assistant Implementation Roadmap

A secure, phased proposal for an Admin and Super Admin AI assistant

Purpose	Plan the AI assistant safely before development begins.
Audience	System owner, Admins, Super Admins, and technical implementation team.
Current decision	Planning only. No application changes are authorised by this document.

Executive Recommendation

Build an app-wide assistant that has controlled, live access to approved data tools rather than unrestricted access to the database. It should understand the full logistics and finance workflow, but every request must pass the same role, branch, and permission rules as the rest of the application.

Recommended first release
An Admin-only, read-only assistant that answers live-data questions, cites source records, and opens the relevant container, invoice, payment, schedule, overhead expense, or report. It must not perform writes in the first release.

Target Operating Model

Admin question	Permission check	Approved live-data tools	Answer with record links	Audit log
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The model never receives database credentials and never runs arbitrary SQL. It requests a limited internal tool; the backend then applies user scope and returns only authorised structured results.

Scope By Stage

Stage	What the assistant may do	What remains prohibited
Release 1	Read live data, analyse, calculate, explain, link to records.	Write, approve, pay, delete, notify, or alter records.
Release 2	Search authorised uploaded documents and	Act on a draft without a human

	prepare reports.	review.
Release 3	Prepare drafts for permitted actions and submit only after confirmation.	Bypass existing approval, payment, officer, or branch controls.
Release 4	Proactive risk and performance monitoring.	Make unexplained accounting or operational decisions.

Phase 0: Business Rules and Governance

Set the authority, budget, data boundaries, and evidence requirements before any AI integration.

Deliverables

- Confirm Admin and Super Admin access for version one.
- Document branch visibility rules and approved questions.
- Set monthly AI usage, retention, and support budgets.
- Define the audit record required for questions, answers, sources, drafts, confirmations, and failures.

Controls

- No sensitive action may be automated.
- Every data result must be attributable to live source records.

Exit criteria

Written policy and acceptance criteria approved by management.

Phase 1: Secure AI Foundation

Create the backend and audit foundation without granting direct database access to a model.

Deliverables

- Admin-only assistant route and chat API.
- Conversation, tool-use, response, and action audit tables.
- Authentication, role, branch, CSRF, rate-limit, and usage-limit enforcement.
- Prompt-injection safeguards and server-side request validation.

Controls

- No raw SQL capability.
- No database credentials, passwords, API keys, or bank secrets sent to the model.

Exit criteria

Security review confirms the assistant cannot exceed the current user's permissions.

Phase 2: Approved Live-Data Tools

Expose carefully designed backend tools that answer real questions from the current application state.

Deliverables

- Dashboard and container summaries.
- Overdue containers, terminal count, delays, documentation, and workflow stage checks.
- Invoices, accounts receivable, payments, schedules, overhead expenses, banks, wallets, and branch comparisons.
- Container timeline and root-cause investigation tools.

Controls

- Each tool applies role, branch, ownership, and record access checks.
- Structured outputs include record IDs and deep links for verification.

Exit criteria

A set of representative management questions returns current, cited answers.

Phase 3: Read-Only Assistant Experience

Deliver the first useful user-facing assistant without operational risk.

Deliverables

- Clean chat screen available only to authorised users.
- Suggested questions for containers, delays, outstanding payments, and branch performance.
- Answer citations and buttons to open the exact source record.
- Clear handling for no data, restricted data, errors, and incomplete records.

Controls

- Answers distinguish facts, calculations, and assumptions.
- The assistant must say when data is incomplete rather than guessing.

Exit criteria

Admins can obtain useful answers and open the evidence behind every significant result.

Phase 4: Document Intelligence

Enable authorised search and summary of uploaded PDFs, images, invoices, bills of lading, PAARs, and Customs documents.

Deliverables

- Text extraction on upload and secure searchable document index.
- Container, branch, document-section, uploader, and permission metadata per indexed item.
- Document answers with filename, page, and source links where available.

Controls

- Document search reuses existing document and branch access control.
- Do not index unsupported or failed uploads silently; show status and retry paths.

Exit criteria

An authorised admin can ask document questions and trace answers to the source file/page.

Phase 5: Reports and Analysis

Turn trusted live data into management-ready reports and explainable risk analysis.

Deliverables

- Monthly revenue, expense, ageing, branch performance, container-delay, and outstanding-payment reports.
- Report preview, source totals, and browser/PDF/Excel export paths.
- Explainable rule-based flags for duplicates, unusual expenses, delayed stages, and missing information.

Controls

- Generated reports are drafts until the user approves export or distribution.
- Anomaly flags are labelled as review prompts, not final accusations or accounting decisions.

Exit criteria

Generated totals reconcile with the existing reports and source records.

Phase 6: Controlled Assisted Actions

Allow the assistant to prepare permitted work while retaining human and server-side control.

Deliverables

- Draft debit notes, payment reminders, schedules, notifications, and management summaries.
- Preview, explicit confirmation, and normal backend validation for every action.
- Action result, user confirmation, and source records written to the audit log.

Controls

- No delete actions in this phase.
- Payment, approval, verification, and officer actions always preserve existing permission checks.

Exit criteria

A confirmed draft action cannot be executed by an unauthorised user or bypass normal workflows.

Phase 7: Proactive Intelligence

Provide scheduled, explainable management insights after the underlying assistant is proven reliable.

Deliverables

- Daily and weekly management briefings.

- Berthing, ageing, delay, PAAR, DO, TDO, Pullout, receivable, and schedule risk summaries.
- Branch comparison and trend narratives with direct links to affected records.

Controls

- Scheduled alerts honour administrator preferences and recipient rules.
- Insights are based on defined thresholds and evidence, not unexplained model speculation.

Exit criteria

Management receives useful prioritised summaries without notification noise or untraceable claims.

Core Safety Requirements

- Server-side role, branch, record, and ownership checks for every AI tool call.
- Immutable AI audit history with user, timestamps, request, answer, sources, tool calls, and confirmed actions.
- Source links and calculations visible in answers so users can validate the result.
- Rate limits, cost controls, retention limits, and monitoring for failed or unusual usage.
- Human confirmation plus normal application validation for all future write actions.
- Automated tests covering permission boundaries, data isolation, tool accuracy, and action confirmation.

First Release Question Set

- How many containers are currently in the terminal?
- Show all overdue containers and their responsible officers.
- Which jobs are delayed at Documentation, Transire, Shipping, Terminal, or Pullout?
- Why is this container delayed?
- Show outstanding invoices, approved schedules awaiting payment, and overdue client balances.
- Compare branch performance this month.
- Show unusual overhead expenses or payments that need review.

Decision gate before development

Approve Phase 0, choose an AI provider and budget, confirm the first-release question list, and identify the Admin/Super Admin access rules. Only then should engineering begin Phase 1.